

CO1-output:

```
Sample Output

Tourist Destinations (Sorted Order):

Araku
GolcondaFort
Hyderabad
NagarjunaSagar
RamojiFilmCity
Vizag
Warangal

Destination Found: Araku

AVL Tree Operations Completed Successfully
```

CO2-output:

```
=== TOURPLAN HOTEL PRICE QUERY ===

Hotels and Prices:

Hotel Paradise -> ₹2500
Green Valley Resort -> ₹1800
Royal Stay -> ₹3200
Ocean View -> ₹1500
Hilltop Inn -> ₹2800
City Comfort -> ₹2100
Sunrise Residency -> ₹1700

Minimum Hotel Price from index 1 to 5: ₹1500

Updating Ocean View price from ₹1500 to ₹1400

New Minimum Hotel Price from index 1 to 5: ₹1400

Segment Tree Query Executed Successfully
```

CO3-output:

```
=== TOURPLAN CITY CONNECTIVITY SYSTEM ===

BFS Traversal: 0 1 2 3 4

DFS Traversal: 0 1 2 4 3

Minimum Spanning Tree:
Hyderabad -> Warangal = 120 km
Warangal -> Vizag = 100 km
Warangal -> Araku = 200 km
Vizag -> Vijayawada = 150 km

Total MST Cost = 570 km
```

CO4-output:

```
=== TOURPLAN PACKAGE RANKING SYSTEM ===
```

```
Popularity Scores Before Sorting:  
[95, 80, 75, 90, 85]
```

```
Popularity Scores After Heap Sort:  
[75, 80, 85, 90, 95]
```

```
Selected Tourist Packages:
```

```
Araku Cost = ₹1500
```

```
Vizag Cost = ₹2500
```

```
Warangal Cost = ₹1000
```

```
LCS Length = 5
```

```
C04 Execution Completed Successfully
```

C05-output:

```
=== TOURPLAN ROUTE RECOMMENDATION SYSTEM ===
```

```
Tourist Route Traversal:
```

```
Hyderabad Warangal Vizag Araku Vijayawada
```

```
Selected Travel Packages:
```

```
Araku -> ₹1500
```

```
Vizag -> ₹2500
```

```
Warangal -> ₹1000
```

```
Optimized Itinerary Days = 7
```

```
C05 Execution Completed Successfully
```

C06-output:

Sample Output

```
===== TOURPLAN SMART TOURISM SYSTEM =====
```

```
Available Tourist Destinations:
```

1. Hyderabad
2. Warangal
3. Vizag
4. Araku
5. Vijayawada

```
Recommended Tourist Route (DFS Traversal):
```

```
Hyderabad Warangal Vizag Araku Vijayawada
```

```
Travel Cost : Rs.4500
```

```
Travel Duration : 7 Hours
```

```
Best Hotels:
```

1. Hotel Green Valley
2. Tourist Residency

```
Optimized Itinerary Generated Successfully
```

```
C06 Execution Completed Successfully
```